Re-analysis Summary Report for Peer-evaluation

Paper ID: **McDevitt_JournPoliEco_2014_yQeR**

Re-analyst's ID: **4o1n4**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
The average plumbing firm whose name begins with A or a number receives… more service complaints than other firms …(p. 909.)

The corresponding article

Here you can find the original article: https://osf.io/6fusy/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/z7mfu/?view_only=277b7d18bb24436eb5cfca69a07eb07a

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
For preprocessing, I centered continuous variables and effect coded the categorical variables (-.5, .5). The data were counts, so I began with poisson regression. I ran a poisson regression of complaints predicted by firm name (A/number or not) controlling for firm age, log employee size, number of names for the same firm, whether or not the firm had multiple names (yes or no), ad spending, whether the firm was in metro Chicago or not, and whether it was on google or not. This revealed a significant relationship between firm name and number of complaints, b = .43, SE = .09, z(2284) = 5.04, p < .001. However, when I computed robust SE (.22, 95% CI = [-.01, .87]), the 95% CI included zero. Also, the deviance of the model was high. Therefore I ran a negative binomial regression, which unlike the poisson regression, does not assume the variance is the same as the mean. This also revealed a significant role of name, such that firms with names starting with A or number had 1.77 times more complaints, b = .57, SE = .22, z(2284) = 2.61, p = .01, 95% CI = [.15, 1.01].

The analyst's conclusion: There is evidence that relative to non-A/non-number plumbing firms, A-name/number-name plumbing firms received more service complaints than other firms. With less control variables, it does seem that A-name/number-name plumbing firms receive significantly more complaints (2.48 X more). Including more control variables, this pattern is attenuated, but still significant– firms with A/number names received 1.77 X more complaints relative to those firms without A/number names.

The analyst's categorization of the result:
The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
Use only the data of Illinois Plumbing Firms in your model. Do not restrict your sample to those firms that serve the metro Chicago area. Do not include the reviews received via yelp.com, Angie's List, and Consumer's Checkbook into your analysis. You should use the overall complaints number instead of complaints per employee.

Here, the analyst reported the following steps and results of the analysis:
I centered continuous variables, and effect coded (-.5, .5) any factors. I log transformed # of employees as the paper did. Given that the data were counts, I ran a poisson regression with firm name (A/number vs. other) predicting number of complaints, with firm age, employee size, number of names, whether they had multiple names, chicago metro area, ad spending, and whether they were on google or not as controls. Since I misunderstood Task 1 and ran many analyses, for Task 2, I simply removed some of the analyses. There is evidence that relative to non-A/non-number plumbing firms, A-name/number-name plumbing firms received more service complaints than other firms. With less control variables, it does seem that A-name/number-name plumbing firms receive significantly more complaints (1.54 X), $b = .43$, $SE = .09$, $t(2284) = 5.09$, $p < .001$. However, using robust standard errors, we find that this relationship is more tenuous, as the 95% CI contains zero, robust $SE = .22$, $p = .054$, 95% CI = [-.008, .87]. This is what inspired me to use the negative binomial regression in Task 1.

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/q8d7y/